

	1	2	3	4	5	6	7	8
A	power_input File: power_input.kicad_sch	boost_charger File: boost_charger.kicad_sch	cap_bank File: cap_bank.kicad_sch	pulse_switch File: pulse_switch.kicad_sch				
B								
C	flyback File: flyback.kicad_sch	sense_gates File: sense_gates.kicad_sch	mcu_status File: mcu_status.kicad_sch					
D								
E								
F								

OmniMarble

Sheet: /
File: omnimarble-driver.kicad_sch

Title: OmniMarble 60V SELV coilgun driver

Size: A3

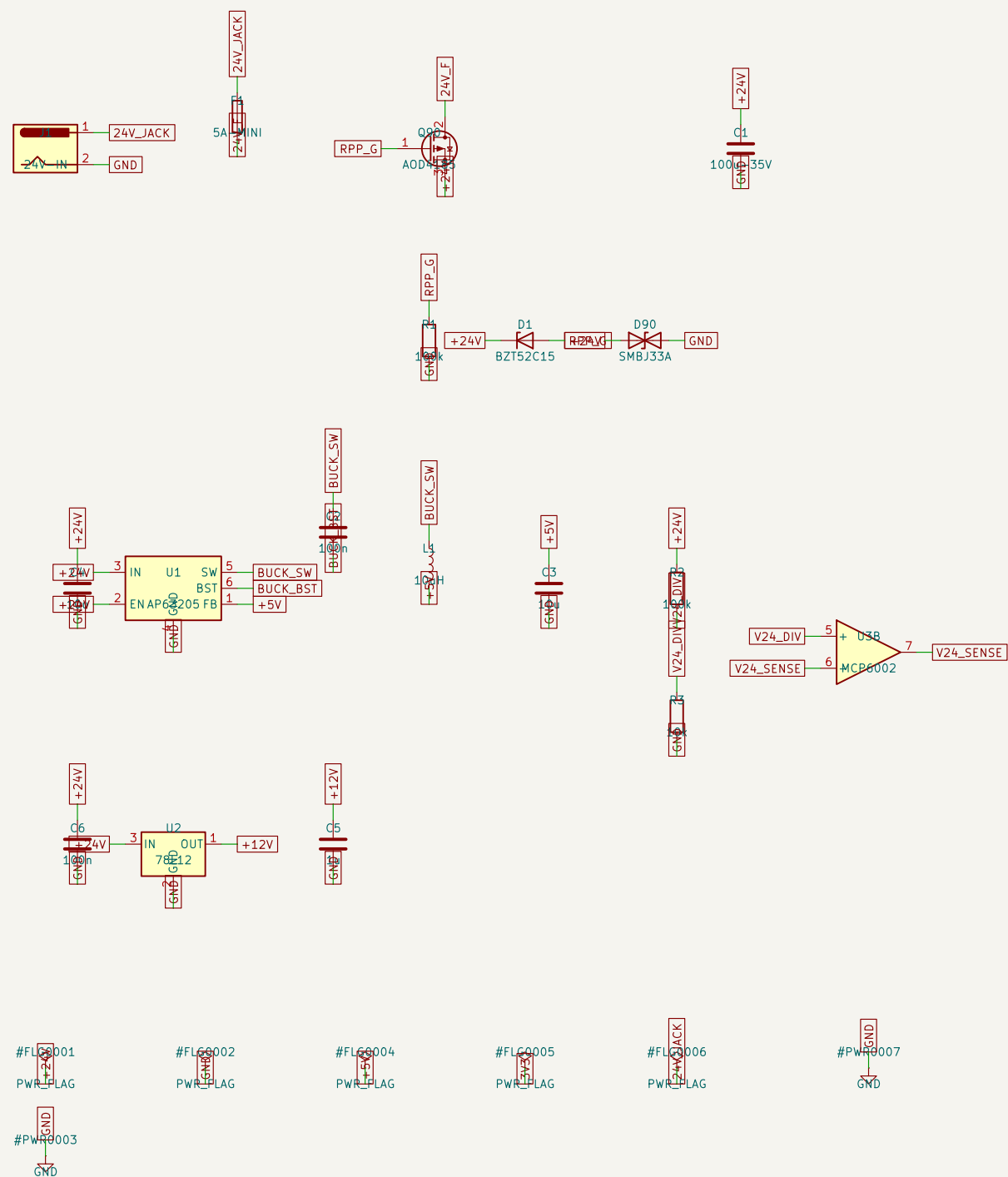
Date:

Rev:

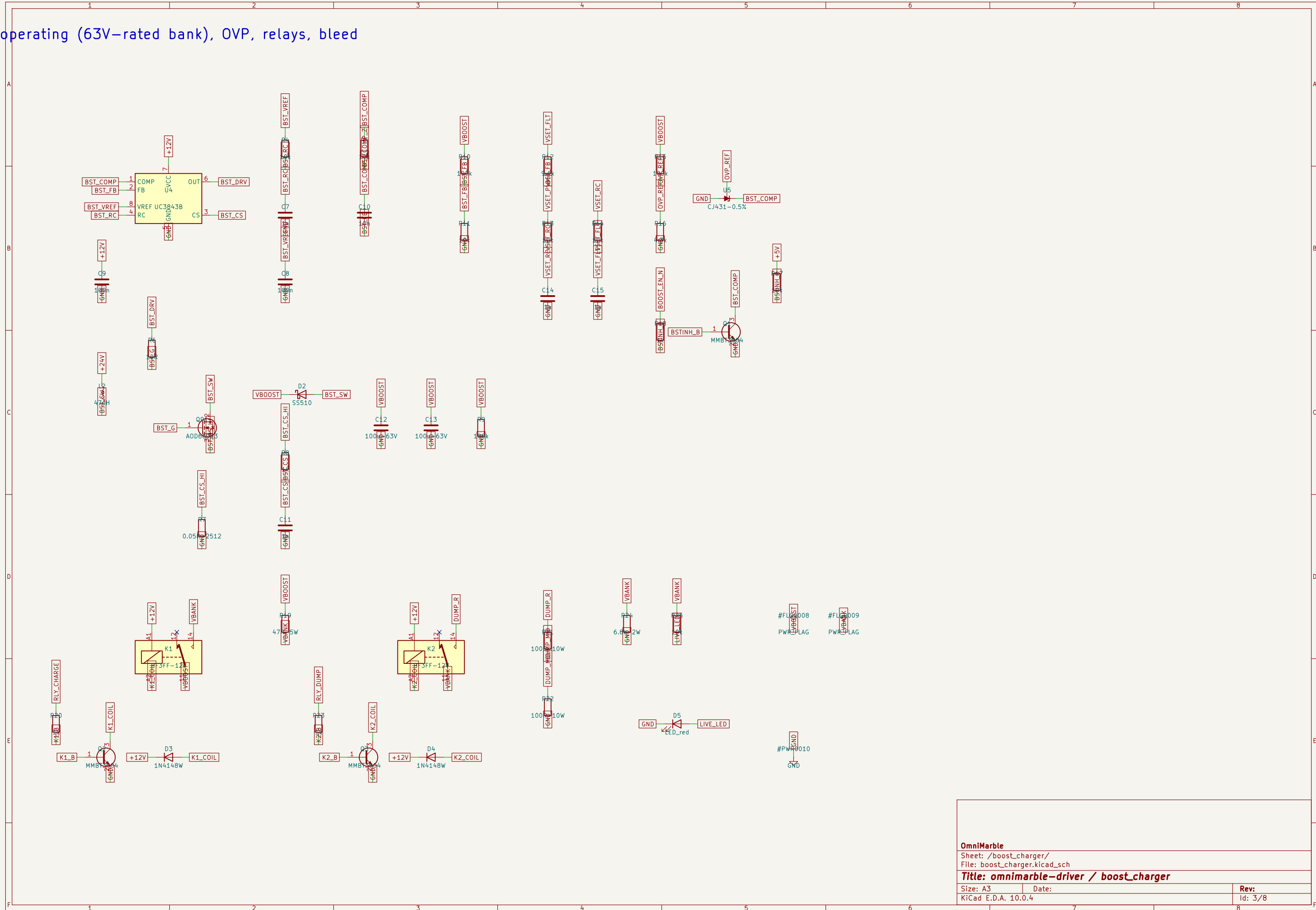
KiCad E.D.A. 10.0.4

Id: 8/8

out, protection, 5V buck, 12V rail

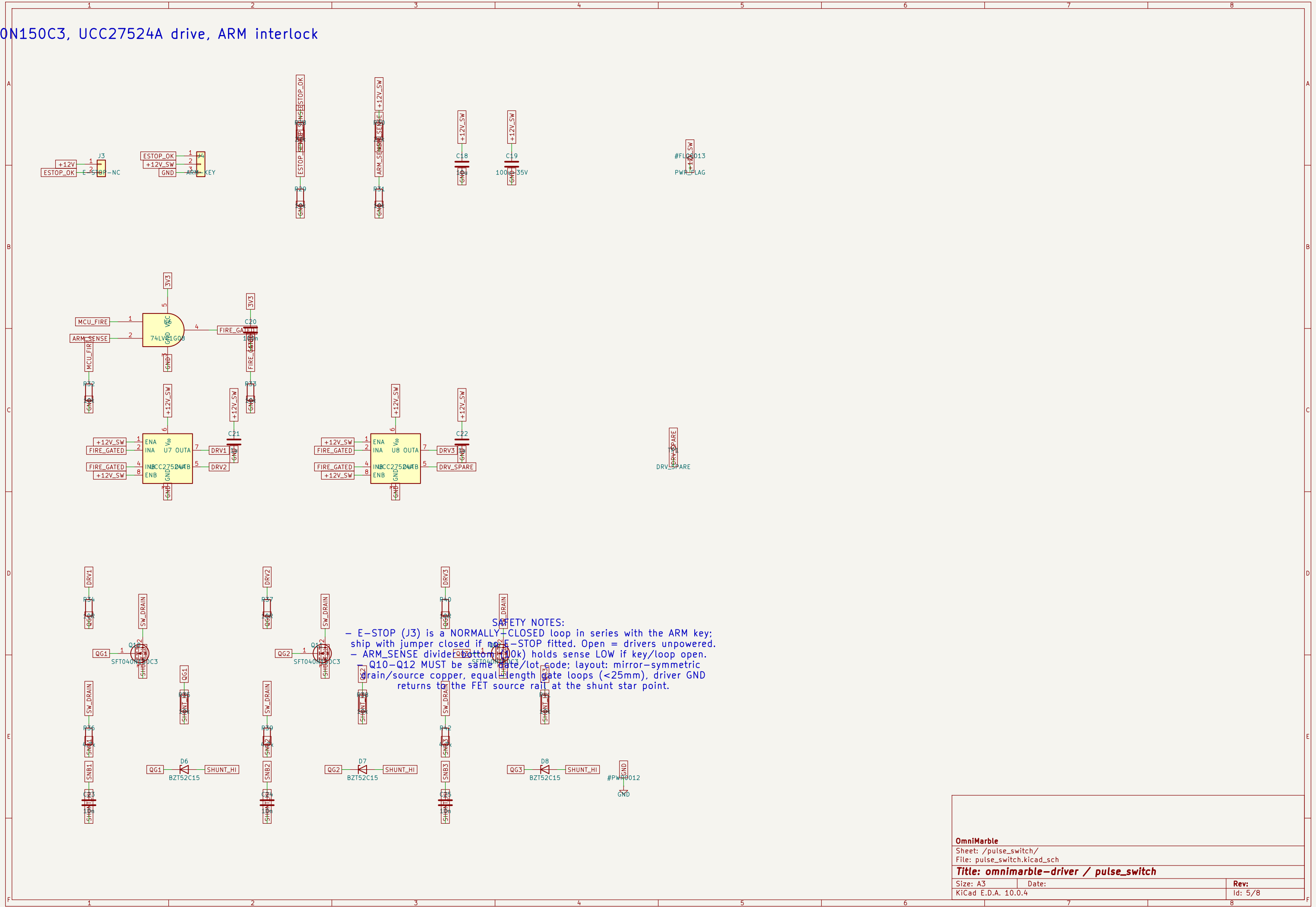


5V operating (63V-rated bank), OVP, relays, bleed

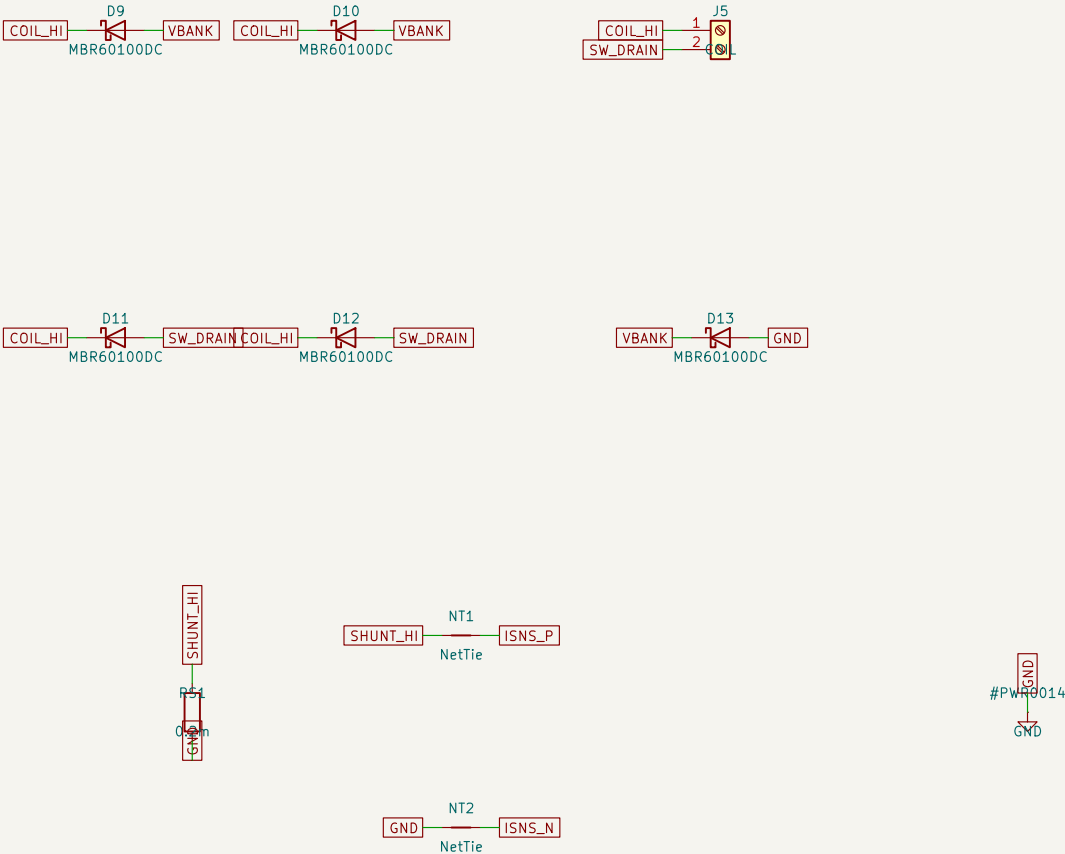


2200µF/63V snap-in (populate per bank table)

T040N150C3, UCC27524A drive, ARM interlock



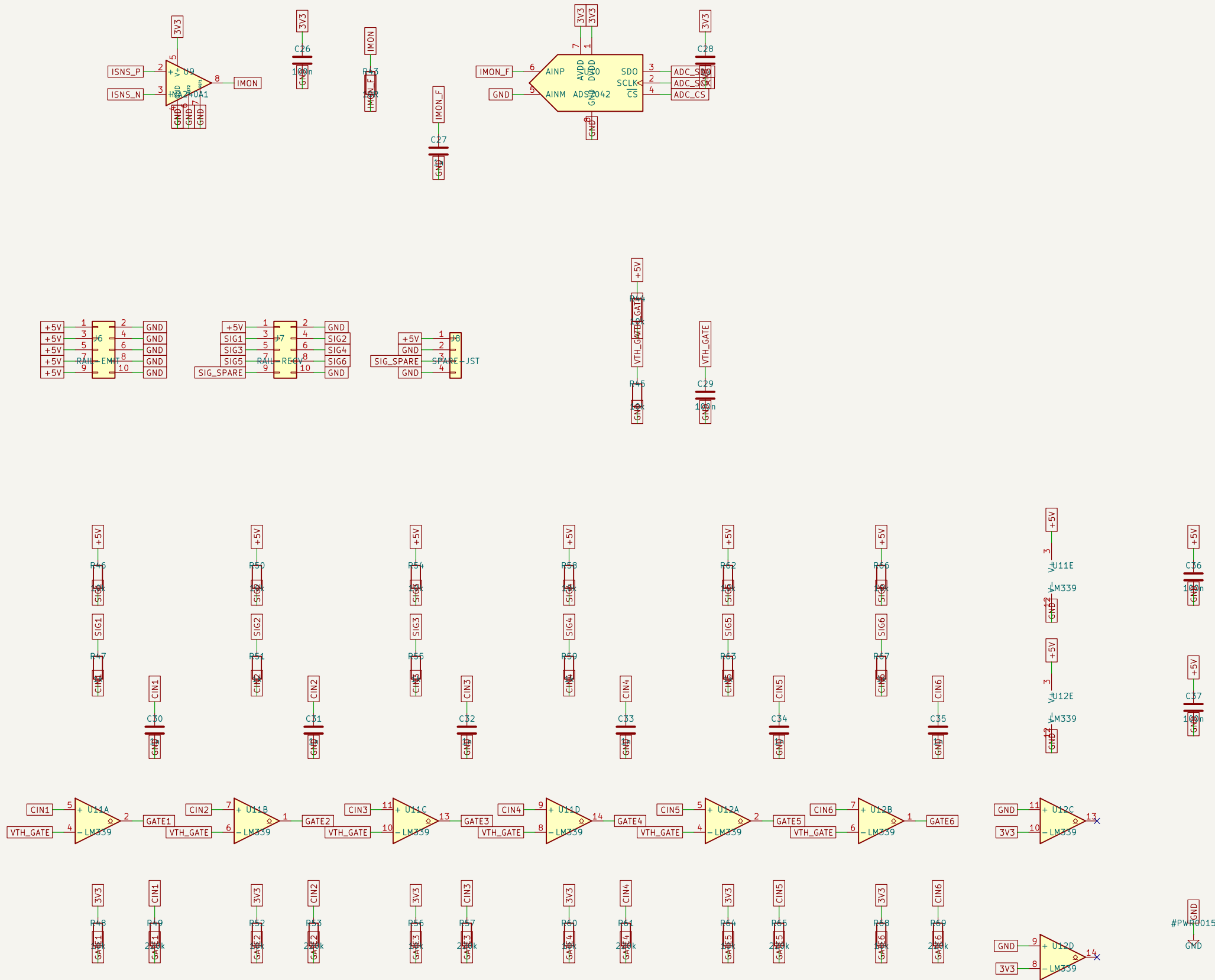
blocking -> COIL -> switch -> shunt -> GND



- validated envelope, swept in calcs.py at 11000uF / 55V:
>= 1 uH AND R_total >= 90 mOhm
coil DC resistance + leads/lugs; keep leads short)
fault: 30T demo coil, 12.406 uH, 110 mOhm total
se can exceed the 600A switch design point – DO NOT FIRE.

OmniMarble		
Sheet: /flyback/ File: flyback.kicad_sch		
Title: omnimarble-driver / flyback		
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 6/8

42 current capture; 6x IR gate comparators



IR GATE TRUTH TABLE:

T -> phototransistor ON -> SIGx ~0V -> comparator OUT LOW
-> phototransistor OFF -> SIGx ~5V -> comparator OUT HIGH
GH = marble in beam. Firmware timestamps the RISING edge.
open-collector outputs pulled to 3V3: Pico-safe levels)

OmniMarble

Sheet: /sense_gates/
File: sense_gates.kicad_sch

Title: omnimarble-driver / sense_gates

Size: A3

Date:

Rev:

KiCad E.D.A. 10.0.4

Id: 7/8

0), MCU-agnostic breakout, relay/status drive

A

B

C

D

E

F

A

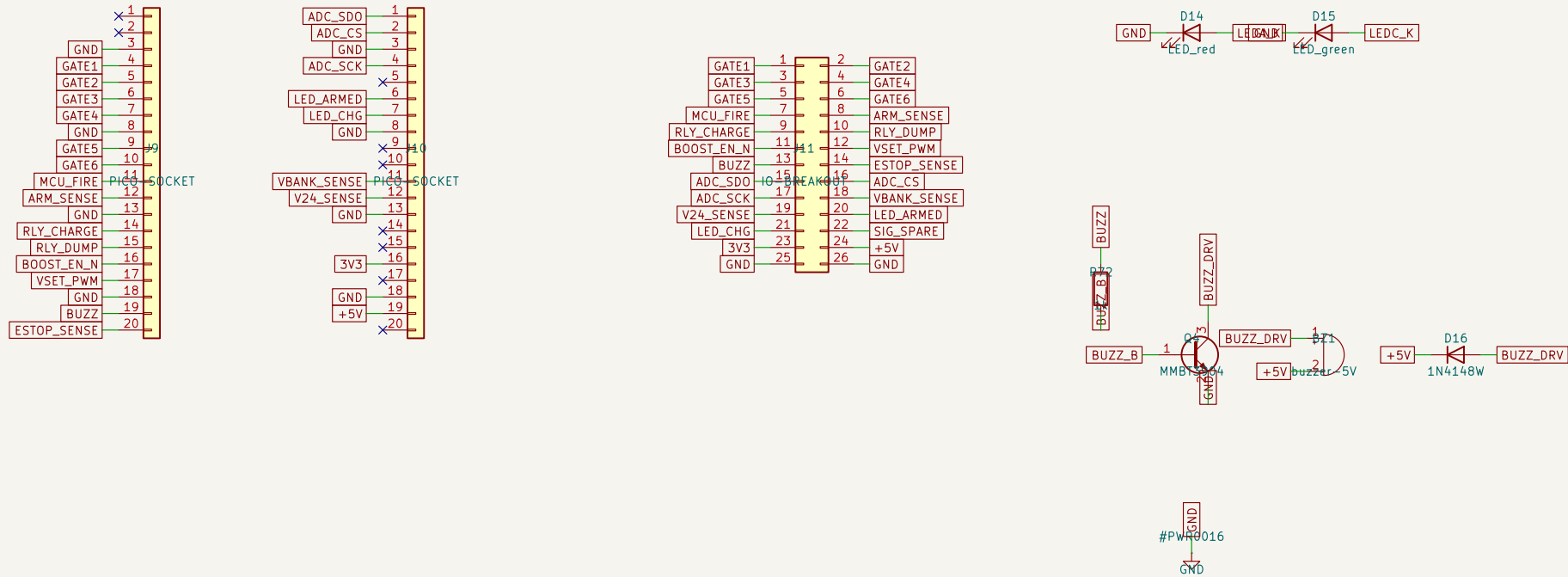
B

C

D

E

F



OmniMarble

Sheet: /mcu_status/
File: mcu_status.kicad_sch

Title: omnimarble-driver / mcu_status

Size: A3

Date:

Rev:

KiCad E.D.A. 10.0.4

Id: 8/8